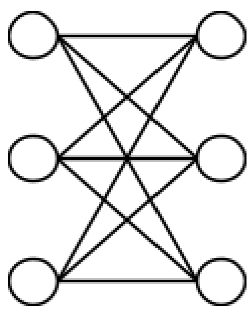
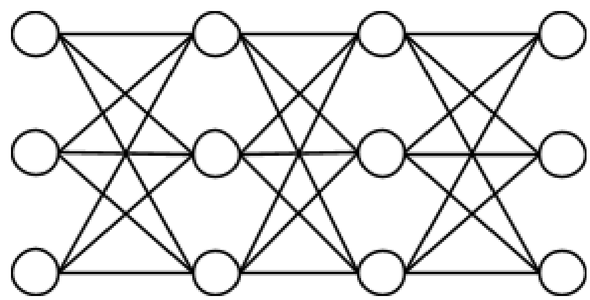
Finding the cop-barricade number C_B for complete staged bipartite graphs

A key example in the pursuit-evasion game cops, robbers, and barricades introduced by Dr. Erin Meger in 2016 is the complete staged bipartite graph, a variant of the complete bipartite graph. A complete staged bipartite graph is composed of d disjoint sets containing k disconnected nodes, completely connected to subsequent sets by edges. The resulting graph is a structure in which nothing can travel directly between members of one set without first visiting members of another set.



complete bipartite graph



staged complete bipartite graph (d=4, k=3)

For the purposes of this informal proof, we will use a staged complete bipartite graph as pictured, with d disjoint sets and k nodes in each set. Let G represent this specific graph throughout this proof. For simplicity, each disjoint set will be referred to as a "stage" of the graph. As a refresher, the rules of standard cops-and-robbers played on any arbitrary graph are as follows:

- The game is played on an undirected, unweighted, connected graph $G = (V, E)$.
- The cops choose their starting vertices first.
- Then the robber places her starting vertex after all cops have been placed.
- The players alternate turns, starting with the cops. On a player's turn, they can move to any adjacent vertex along an edge, or they can choose to stay put on their current vertex.
- Both cops and robbers move at the same speed, traveling a maximum of one edge per turn. All cops individually move or choose to remain stationary on the cop turn.
- The cops win if a cop moves onto the same vertex as the robber.

- The robber wins if she evades capture indefinitely, meaning she can always find a valid adjacent vertex that a cop is not moving to on the next turn.
- The cop number $C(G)$ is the minimum number of cops required to guarantee the capture of one robber on that specific graph G. If $C(G) = 1$ then G is "cop-win".

Lemma 1: With standard pursuit-evasion cops-and-robbers rules, the cop number of G, denoted $C(G)$, is 2, regardless of the values of d or k.

Proof: Assert $d \geq 2$ and $k \geq 2$; otherwise, it is no longer a staged complete bipartite graph. Assuming both the cop C and robber R are playing optimally, the gameplay is as follows:

➢ Let C start in some arbitrary stage X_i.
➢ Let R start in some X_{i+j} where $j \geq 2$.
➢ Eventually, C will get within 1 stage of R.
➢ On the next turn, R will move to $X_{i'}$ and is not adjacent to C.
➢ C will move to $X_{i\pm1}$.
➢ R will follow, evading capture indefinitely.

At the end of every robber move, the robber occupies a stage not occupied by C and adjacent to C's stage. Since C moves first and R always has a neighboring stage available, this invariant is maintained forever. Therefore, $C(G) > 1$. If we increase the number of cops to 2 and place them optimally on G, the gameplay is as follows:

➢ Place both cops on the center-most stage. If d is an odd number, they will be placed on $X_{\lceil \frac{d}{2} \rceil}$; if even, they will be placed individually on $X_{\frac{d}{2}}$ and $X_{\frac{d}{2}+1}$.
➢ R will begin in some X_j without loss of generality.
➢ Let $0 \leq j < \dfrac{d}{2}$.
➢ The cops C_1 and C_2 will move towards R, staying 1 stage apart. That is, they move in tandem.
➢ (If they started started on the same vertex, C_2 will remain stationary while C_1 moves ahead; then they will move in tandem.)

- ➢ R can never move past a cop because every stage is completely joined to its adjacent stage.
- ➢ Eventually, they will push R to one end of the graph.
- ➢ R cannot remain stationary or she will be captured by C_1, but she cannot move to any adjacent vertex or she will be captured by C_2. There are no other moves; the cops will win.

Regardless of d or k, at least 2 cops are required to guarantee the robber's capture. $C(G) = 2$.

This result is self-evident. But how does the barricade mechanic change game behaviour? As described in Dr. Meger's 2016 Masters of Mathematics thesis, the rules of cops, robbers, and barricades are as follows:

- CRB follows all rules of standard cops-and-robbers, with the addition of barricade mechanics.
- CRB is played on a reflexive graph G. It is permissible for a player to remain on the same vertex.
- The robber, on her turn, is allowed to move to any adjacent vertex, or instead of moving is permitted to build one barricade on an adjacent vertex. There are other variations of barricade rules, but for the purposes of this proof, the robber is playing the "local variant" — that is, she can build one barricade per turn, instead of moving, and her barricaded vertex must be adjacent to her current location.
- A barricade forbids any player from entering that vertex, essentially deleting it from the graph.
- The barricade-cop number of a graph, $C_B(G)$, is the minimum number of cops required to guarantee capture. If $C_B(G) = 1$, then G is "barricade-cop-win".

Barricades can only help R because she still has vanilla cops-and-robbers methods available to her, so $C_B(G) \geq 2$. This is supported by Lemma 5 on page 19 of the thesis: $\forall G$, $C(G) \leq C_B(G)$.

Lemma 2: A k-barricade wall succeeds iff every threatening cop is at a distance $\geq k + 2$ from R at the moment the wall begins.

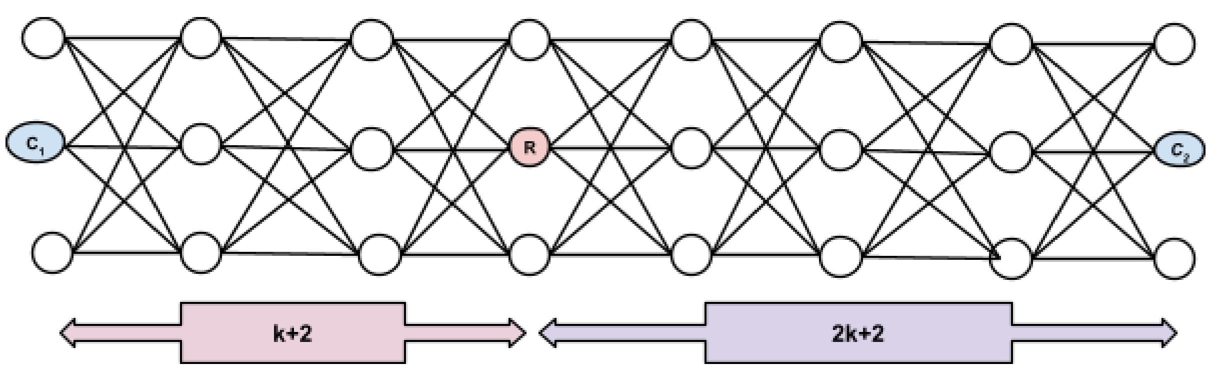
Proof: Let R freeze in round t to build a barricade wall in her adjacent stage with some cop C placed j stages away from her on that side as of round t. After C's mth move, he is at distance $j-m$. He will enter the wall stage at distance 1, and he will always be able to stand on an unbarricaded vertex because every wall-stage vertex is adjacent to his previous stage, regardless of what node he is on.

If $j=k+1$, then after C's kth move, he will stand in the wall stage on the last unbarricaded vertex. R will be unable to place the last barricade, and she will be captured. If $j \geq k+2$, then she can place her final barricade before the cop reaches her.

Corollary 1: It follows that any "blind spot" in the graph — that is, the very end stages of G, X_1 and X_d — unoccupied by a cop is still acceptable for the cops if the closest cop C is not more than $k+1$ stages away from X_1 or X_d.

Lemma 3: Cops can be placed at a maximum of $3k+3$ stages apart before the robber can win.

Proof: Let C_1 and C_2 approach R from either side, $k+2$ and $2k+2$ stages apart, respectively, from the set occupied by R.
R will build a wall on the side of C_1 and let C_2 approach her from the other side.



The distance between C_1 and C_2 is, then,
$k+2+2k+2=3k+4$.

Let j_L, j_R be her distances to the nearest cop on each side, so $j_L + j_R = 3k + 4$.
R will begin building the second wall at round $k + 1$, by which point C_2 has already advanced k stages. Applying Lemma 2 to that wall at its start time gives $j_R - k \geq k + 2$, so $j_R \geq 2k + 2$.
C_2 is now $k + 2$ stages away from R when R begins building the second wall. This wall costs R $k + 1$ turns, and with one turn left, C_2 will be closed off and R will win. Therefore, if $j_L + j_R \geq 3k + 4$, R will win.
Thus, if the distance between C_1 and C_2 is no more than $3k + 3$ stages, R will be captured. This conclusion generalizes to n cops because k and d can be arbitrary.

With Lemmas 1 through 3 and , we can begin to construct a general formula for $C_B(G)$ as a function of k (the number of vertices per set), d (the number of stages), and n (the number of cops):

General formula for $C_B(G)$ in complete staged bipartite graphs

Let m be the distance between an arbitrary C and R. Walking from stage X_1 to stage X_d is $d - 1$ steps. There are $n - 1$ gaps between n cops at a length of $3k + 3$ each, at most. Corollary 1 gives us that a cop can have a blind spot of at most $k + 1$.

So, if the cops are distributed evenly enough throughout the graph such that the total length of the graph is not greater than the maximum distance between all of the cops, then the cops will win. This can be proven algebraically:

$$d - 1 \leq (k + 1) + (n - 1)(3k + 3) + (k + 1)$$
$$d \leq (k + 1) + (n - 1)(3k + 3) + (k + 1) + 1$$
$$d \leq 2k + 3 + (n - 1)(3k + 3)$$

From here, we solve for n as a function of d and k.

$$d - 2k - 3 \leq (n - 1)(3k + 3)$$
$$d - 2k - 3 \leq 3kn + 3n - 3k - 3$$
$$d - 2k - 3 + 3k + 3 \leq 3kn + 3n$$
$$d + k \leq 3kn + 3n$$
$$d + k \leq 3n(k + 1)$$

$\frac{d+k}{k+1} \leq 3n$

$\frac{d+k}{3k+3} \leq n$

We must take the ceiling of this fraction because $C(G) \leq C_B(G)$.

$$n \geq \left\lceil \frac{d+k}{3k+3} \right\rceil$$

So n cops is sufficient for the cop team to win a game of cops, robbers, and barricades played on a complete staged bipartite graph *iff* the inequality holds. Therefore:

$$C_B(G_{d,k} = \max\Big(C(G), \left\lceil \frac{d+k}{3k+3} \right\rceil\Big)$$

And because from Lemma 1 we have that for any complete staged bipartite graph, $C(G) = 2$:

$$C_B(G_{d,k} = \max\Big(2, \left\lceil \frac{d+k}{3k+3} \right\rceil\Big)$$

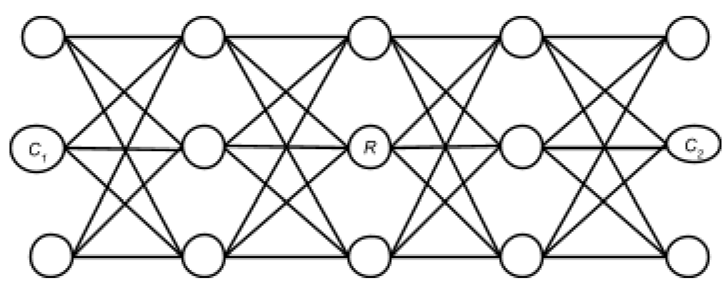
ROUGH WORK/INCORRECT/QUESTIONABLE:

If every other vertex is occupied by a cop, R can never win, so for any graph G, $2 \leq C_B(G) < d$. R's goal thus changes — if she barricades every vertex of a stage, the graph splits into two pieces. If all cops are on one side of the graph and the robber is safely on the other side, R will win because there is no way for a cop to reach her.

If R is on one end of G with one cop C approaching from the other, R needs to build k barricades to prevent C from reaching her. If the initial distance between C and R is greater than k, then the robber will win.

As discussed, if cops C_1 and C_2 are placed on both ends of G with R in the middle stage, R must build $2k$ barricades to cut off both cops. However, the primary winning strategy for R is to build k barricades on one side, to block C_1, and perpetually evade C_2 on the other.

Since R chooses her starting position after C_1 and C_2 have already been placed, and all players make optimal decisions, she will always maximize her starting distance from both cops. Barricades change the optimal cop placement from centered to end-anchored, because clustering lets R run to the far side and wall herself off. Therefore, the optimal positions for C_1 and C_2 are X_1 and X_d, so R's ideal starting distance is $\lfloor \frac{d-1}{2} \rfloor$. R has k moves before she is captured by either C_1 or C_2. If she completely blocks off one side, then she can perpetually evade the remaining cop, resulting in a trivial win.



optimal starting positions for R, C_1, and C_2 on $d = 5$, $k = 3$. R will be captured.
note that this is one of multiple optimal starting positions; alternatives exist.

Thus, the win condition becomes: R will be captured by C_1 and C_2 iff $k \geq \left\lfloor \frac{d-1}{2} \right\rfloor$. However, this conclusion does not generalize to $n \geq 3$ — R can't perpetually evade if there are two cops on her section of the graph; the two cops will employ the strategy seen in Lemma 1, guaranteeing her capture.

Lemma 2: A k-barricade wall succeeds iff every threatening cop starts at a distance \gek $+ 2$.
Proof: Let R freeze to build a barricade wall in her adjacent stage with some cop C_a placed j stages away from her on that side. After C_a's mth move, he is at distance $j - m$. He will enter the wall stage at distance 1, and he will always be able to stand on an unbarricaded vertex because every wall-stage vertex is adjacent to his previous stage, regardless of what node he is on.

If $j = k + 1$, then after C_a's kth move, he will stand in the wall stage on the last unbarricaded vertex. R will be unable to place the last barricade, and she will be captured. If $j \geq k + 2$, then she can place her final barricade before the cop reaches her.

The non-wall side has the same bound. C_b approaches R while she is frozen, placing barricades on the opposite side, but will not get within 1 stage of her before she is able to evade, provided he started at least $k + 2$ stages away from her.

?????

R only needs to seal herself off from the nearest cop and evade the rest indefinitely, so C_1, ..., C_n need to position themselves so that no matter where R sits, the inequality holds — otherwise, she will get enough of a head start so she can block off an entire set.

With n cops spread across X_1 through X_d, evenly spaced interior ones create $n - 1$ gaps between consecutive cops, each of length $\left\lfloor \frac{d-1}{n-1} \right\rfloor$. R will always pick the middle of the biggest gap, giving her a head-start distance of $\left\lfloor \frac{d-1}{2(n-1)} \right\rfloor$.

i think thats wrong